YouTube Scam Comments Detection Scheme Using Machine Learning

A PROJECT REPORT

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ABSTRACT

This paper proposes a technique to detect spam comments on YouTube, which have recently seen tremendous growth. YouTube is running its own spam blocking system but continues to fail to block them properly. Therefore, we examined related studies on YouTube spam comment screening and conducted classification experiments with six different machine learning techniques Naïve Bayes and two ensemble models (Ensemble with hard voting, Ensemble with soft voting) combining these techniques in the comment data from popular music videos - Psy, Katy Perry, LMFAO, Eminem and Shakira

These tools will block malicious links, however they cannot protect the user in real-time as early as possible. Thus, industries and researchers have applied completely different approaches to form spam free social network platform. The survey for the spam comments detection methodology has been carried out using four Artificial Intelligence estimations . With the use of Neural Network, we can achieve an exactness of 91.65% and beat the present course of action by around 18%. The most notable AI procedures (Bayesian portrayal, k-NN, ANNs, SVMs) and of their suitability to the issue of spam.

CHAPTER 1

INTRODUCTION

1.1 GENERAL

- In the previous years, informal online communities like Face book and YouTube have become progressively common platform in an individual person's day to day life. People use social media as a virtual community platform to stay in touch with friends and family and to also share thoughts and ideas in blogs. Due to this developing pattern, these platforms pull in an enormous number of clients and are easy targets for spammers. YouTube has become the most well-known informal community among youngsters. For example, many makeup tutorials have been started by bloggers who are referred to as "beauty guru" or "beauty influencers" in which majority of the audiences are teenage girls. These days, 200 million clients produce 400 million new YouTube content (videos) every day. This extensive environment provided by YouTube also creates an opportunity for spammers to create irrelevant content directed to users. These irrelevant or unsolicited messages are aimed to attack users by luring them into clicking links to view malicious sites containing malware, phising and scams. One of the most highlighted features of YouTube is the comments section below every video posted by a user YouTube, the world's largest video sharing site, was founded in 2005 and acquired by Google in 2006. YouTube has grown tremendously as a video content platform, with the recent shift in online content to video. At present, more than 400 hours of video are uploaded and 4.5 million videos are watched every minute on YouTube . It is easy for users to watch and upload videos without any restrictions. This great accessibility has increased the number of personal media, and some of them have become online influencers. YouTube creators can monetize if they have more than 1,000 subscribers and 4,000 hours of watch time for the last 12 months. Accordingly, spam comments are being created to promote their channels or videos in popular videos. Some creators closed the comment function due to aggression such as political comments, abusive speech, or derogatory comments not related to their videos feature allows users to share opinions and ideas. In this project, the prediction of the spam comments present in the comments section of Youtube videos using the concept called machine learning, it is also known as subset of artificial intelligence, is done. Supervised learning approach depends on a very large number of labelled datasets. .The proposed classification algorithm (Logistic Regression) is used in order to predict the spam comment. The purpose of project is to introduce briefly the techniques of machine learning and to outline the prediction technique. Being much more superior to the conventional data analysis techniques, machine learning can open a new opportunity to explore and increase the prediction accuracy. Spam remarks are regularly completely immaterial to the given video and

are normally created via mechanized bots camouflaged as a client. The comments section is target by spammers to post completely irrelevant messages, comments, links and ideas. AI is the strategy for extraction, changing, stacking and anticipating the significant data from enormous information to remove a few examples and furthermore change it into justifiable structure for additional utilization. Grouping and expectation are two sorts of dissecting information which portray principal classes of information and forecast of patterns in future information. The noxious spam remarks will ruin the positive perspective of the contents present in the videos posted. The contingency for anticipating the spam remarks has started but has yet not been concluded and built up for an exact forecast of spam remarks.

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CHAPTER 1

INTRODUCTION

1.1 MACHINE LEARNING – OVERVIEW

Machine learning is a very hot topic for many key reasons, and because it provides the ability to automatically obtain deep insights, recognize unknown patterns, and create high performing predictive models from data, all without requiring explicit programming instructions.

This high level understanding is critical if ever involved in a decision-making process surrounding the usage of machine learning, how it can help achieve business and project goals, which machine learning techniques to use, potential pitfalls, and how to interpret the results.

1.2 MACHINE LEARNING – TASKS

The most common machine learning tasks that one may come across while trying to solve a machine learning problem. Under each task are also listed a set of machine learning methods that could be used to resolve these tasks. Please feel free to comment/suggest if I missed mentioning one or more important points.

Following are the key machine learning tasks briefed later in this article:

- Feature selection
- Regression
- Classification
- Clustering
- Multivariate querying
- Density estimation
- Dimension reduction
- Testing and matching

Following are top 8 most common machine learning tasks that one could come across most frequently while solving an advanced analytics problem:

Feature Selection: Feature selection is one of the critical tasks which would be used when building machine learning models. Feature selection is important because selecting right features would not only help build models of higher accuracy but also help achieve objectives related to building simpler models, reduce overfitting etc. The following are some of the techniques which could be used for feature selection:

Filter methods which helps in selecting features based on the outcomes of statistical tests. The following are some of the statistical tests which are used:

- Pearson's correlation
- Linear discriminant analysis (LDA)
- Analysis of Variance (ANOVA)
- Chi-square tests

Wrapper methods which helps in feature selection by using a subset of features and determining the model accuracy. The following are some of the algorithms used:

- Forward selection
- Backward elimination

Recursive feature elimination

Regularization techniques which penalizes one or more features appropriately to come

up with most important features. The following are some of the algorithms used:

- LASSO (L1) regularization
- Ridge (L2) regularization

Regression: Regression tasks mainly deal with estimation of numerical values (continuous variables). Some of the examples include estimation of housing price, product price, stock price etc. Some of the following ML methods could be used for solving regressions problems:

- Kernel regression *(Higher accuracy)*
- Gaussian process regression *(Higher accuracy)*
- Regression trees
- Linear regression
- Support vector regression
- LASSO

Classification: Classification tasks is simply related with predicting a category of a data (discrete variables). One of the most common example is predicting whether or not an email if spam or ham. Some of the common use cases could be found in the area of healthcare such as whether a person is suffering from a particular disease or not. It also has its application in financial use cases such as determining whether a transaction is fraud or not. The ML methods such as following could be applied to solve classification tasks:

- Kernel discriminant analysis *(Higher accuracy)*
- K-Nearest Neighbors *(Higher accuracy)*
- Artificial neural networks (ANN) *(Higher accuracy)*
- Support vector machine (SVM) *(Higher accuracy)*
- Random forests *(Higher accuracy)*
- Decision trees
- Boosted trees
- Logistic regression
- naive Bayes

DEEP LEARNING

Clustering: Clustering tasks are all about finding natural groupings of data and a label associated with each of these groupings (clusters). Some of the common example includes customer segmentation, product features identification for product roadmap. Some of the following are common ML methods:

- Mean-shift (*Higher accuracy)*
- Hierarchical clustering
- K-means
- Topic models

Multivariate querying: Multivariate querying is about querying or finding similar objects. Some of the following ML methods could be used for such problems:

- Nearest neighbors
- Range search
- Farthest neighbors

Density estimation: Density estimation problems are related with finding likelihood or frequency of objects. In probability and statistics, density estimation is the construction of an estimate, based on observed data, of an unobservable underlying probability density function. Some of the following ML methods could be used for solving density estimation tasks:

- Kernel density estimation *(Higher accuracy)*
- Mixture of Gaussians
- Density estimation tree

Dimension reduction: Dimension reduction is the process of reducing the number of random variables under consideration, and can be divided into feature selection and feature extraction. Following are some of ML methods that could be used for dimension reduction:

- Manifold learning/KPCA *(Higher accuracy)*
- Principal component analysis
- Independent component analysis

- Gaussian graphical models
- Non-negative matrix factorization
- Compressed sensing.

Testing and matching: Testing and matching tasks relates to comparing data sets. Following are some of the methods that could be used for such kind of problems:

- Minimum spanning tree
- Bipartite cross-matching

N-point correlation

1.3 MACHINE LEARNING SYSTEM CLASSIFICATION

Although supervised and unsupervised learning are two of the most widely accepted machine learning methods by businesses today, there are various other machine learning techniques. Following is an overview of some of the most accepted ML methods

SUPERVISED LEARNING

These algorithms are trained using labeled examples, in different scenarios, as an input where the desired outcome is already known. An equipment, for instance, could have data points such as "F" and "R" where "F" represents "failed" and "R" represents "runs".

A learning algorithm will receive a set of input instructions along with the corresponding accurate outcomes. The learning algorithm will then compare the actual outcome with the accurate outcome and flag an error, if there is any discrepancy. Using different methods, such as regression, classification, gradient boosting, and prediction, supervised learning uses different patterns to proactively predict the values of a label on extra unlabeled data. This method is commonly used in areas where historical data is used to predict events that are likely to occur in the future.

UNSUPERVISED LEARNING

This method of ML finds its application in areas were data has no historical labels. Here, the system will not be provided with the "right answer" and the algorithm should identify what is being shown. The main aim here is to analyze the data and identify a pattern and structure within the available data set. Transactional data serves as a good source of data set for unsupervised learning.

For instance, this type of learning identifies customer segments with similar attributes and then lets the business to treat them similarly in marketing campaigns. Similarly, it can also identify attributes that differentiate customer segments from one another. Either ways, it is about identifying a similar structure in the available data set. Besides, these algorithms can also identify outliers in the available data sets.

Some of the widely used techniques of unsupervised learning are

- k-means clustering
- self-organizing maps
- value decomposition
- mapping of nearest neighbor.

SEMI-SUPERVISED LEARNING

This kind of learning is used and applied to the same kind of scenarios where supervised learning is applicable. However, one must note that this technique uses both unlabeled and labeled data for training. Ideally, a small set of labeled data, along with a large volume of unlabeled data is used, as it takes less time, money and efforts to acquire unlabeled data. This type of machine learning is often used with methods, such as regression, classification and prediction.

REINFORCEMENT LEARNING

This is mainly used in navigation, robotics and gaming. Actions that yield the best rewards are identified by algorithms that use trial and error methods. There are three

major components in reinforcement learning, namely, the agent, the actions and the environment.

1.4 MACHINE LEARNING APPLICATIONS

As we move forward into the digital age, one of the modern innovations we've seen is the creation of Machine Learning. This incredible form of artificial intelligence is already being used in various industries and professions. For Example, Image and Speech Recognition, Medical Diagnosis, Prediction, Classification, Learning Associations, Statistical Arbitrage, Extraction, Regression. Today we're looking at all these Machine Learning Applications in today's modern world.

Image Recognition

It is one of the most common machine learning applications. There are many situations where you can classify the object as a digital image. For digital images, the measurements describe the outputs of each pixel in the image.

In the case of a black and white image, the intensity of each pixel serves as one measurement. So if a black and white image has N*N pixels, the total number of pixels and hence measurement is N2.

In the colored image, each pixel considered as providing 3 measurements of the intensities of 3 main color components i.e. RGB. So N*N colored image there are 3 N2 measurements.

For face detection – The categories might be face versus no face present. There might be a separate category for each person in a database of several individuals.
For character recognition – We can segment a piece of writing into smaller images, each containing a single character. The categories might consist of the 26 letters of the English alphabet, the 10 digits, and some special characters.

Speech Recognition

Speech recognition (SR) is the translation of spoken words into text. It is also known as “automatic speech recognition” (ASR), “computer speech recognition”, or “speech to text” (STT).

In speech recognition, a software application recognizes spoken words. The measurements in this Machine Learning application might be a set of numbers that represent the speech signal. We can segment the signal into portions that contain distinct words or phonemes. In each segment, we can represent the speech signal by the intensities or energy in different time-frequency bands.
Although the details of signal representation are outside the scope of this program, we can represent the signal by a set of real values.

Speech recognition, Machine Learning applications include voice user interfaces. Voice user interfaces are such as voice dialing, call routing, demotic appliance control. It can also use as simple data entry, preparation of structured documents, speech-to-text processing, and plane.

Medical Diagnosis

ML provides methods, techniques, and tools that can help in solving diagnostic and prognostic problems in a variety of medical domains. It is being used for the analysis of the importance of clinical parameters and of their combinations for prognosis, e.g. prediction of disease progression, for the extraction of medical knowledge for outcomes research, for therapy planning and support, and for overall patient management. ML is also being used for data analysis, such as detection of regularities in the data by appropriately dealing with imperfect data, interpretation of continuous data used in the Intensive Care Unit, and for intelligent alarming resulting in effective and efficient monitoring.

It is argued that the successful implementation of ML methods can help the integration of computer-based systems in the healthcare environment providing opportunities to facilitate and enhance the work of medical experts and ultimately to improve the

efficiency and quality of medical care.

In medical diagnosis, the main interest is in establishing the existence of a disease followed by its accurate identification. There is a separate category for each disease under consideration and one category for cases where no disease is present. Here, machine learning improves the accuracy of medical diagnosis by analyzing data of patients.

The measurements in this Machine Learning applications are typically the results of certain medical tests (example blood pressure, temperature and various blood tests) or medical diagnostics, presence/absence/intensity of various symptoms and basic physical information about the patient (age, sex, weight etc.). On the basis of the results of these measurements, the doctors narrow down on the disease inflicting the patient.

Statistical Arbitrage

In finance, statistical arbitrage refers to automated trading strategies that are typical of a short-term and involve a large number of securities. In such strategies, the user tries to implement a trading algorithm for a set of securities on the basis of quantities such as historical correlations and general economic variables. These measurements can be cast as a classification or estimation problem. The basic assumption is that prices will move towards a historical average.
We apply machine learning methods to obtain an index arbitrage strategy. In particular, we employ linear regression and support vector regression (SVR) onto the prices of an exchange-traded fund and a stream of stocks. By using principal component analysis (PCA) in reducing the dimension of feature space, we observe the benefit and note the issues in the application of SVR. To generate trading signals, we model the residuals from the previous regression as a mean reverting process.

In the case of classification, the categories might be sold, buy or do nothing for each security. I the case of estimation one might try to predict the expected return of each security over a future time horizon. In this case, one typically needs to use the estimates of the expected return to make a trading decision (buy, sell, etc.)

Learning Associations

Learning association is the process of developing insights into various associations between products. A good example is how seemingly unrelated products may reveal an association to one another. When analyzed in relation to buying behaviors of customers.

One application of machine learning- Often studying the association between the products people buy, which is also known as basket analysis. If a buyer buys 'X', would he or she force to buy 'Y' because of a relationship that can identify between them? This leads to the relationship that exists between fish and chips etc.

when new products launch in the market a Knowing these relationships it develops a new relationship. Knowing these relationships could help in suggesting the associated product to the customer. For a higher likelihood of the customer buying it, it can also help in bundling products for a better package.

This learning of associations between products by a machine is learning associations. Once we found an association by examining a large amount of sales data, Big Data analysts. It can develop a rule to derive a probability test in learning a conditional probability.

Classification

Classification is a process of placing each individual from the population under study in many classes. This is identified as independent variables.

Classification helps analysts to use measurements of an object to identify the category to which that object belongs. To establish an efficient rule, analysts use data. Data consists of many examples of objects with their correct classification.

For example, before a bank decides to disburse a loan, it assesses customers on their ability to repay the loan. By considering factors such as customer's earning, age, savings and financial history we can do it. This information is taken from the past data of the loan. Hence, Seeker uses to create a relationship between customer attributes and related risks.

Prediction

Consider the example of a bank computing the probability of any of loan applicants faulting the loan repayment. To compute the probability of the fault, the system will first need to classify the available data in certain groups. It is described by a set of rules prescribed by the analysts.

Once we do the classification, as per need we can compute the probability. These probability computations can compute across all sectors for varied purposes

The current prediction is one of the hottest machine learning algorithms. Let's take an example of retail, earlier we were able to get insights like sales report last month / year / 5-years / Diwali / Christmas.

Extraction

Information Extraction (IE) is another application of machine learning. It is the process of extracting structured information from unstructured data. For example, web pages, articles, blogs, business reports, and e-mails. The relational database maintains the output produced by the information extraction.

The process of extraction takes input as a set of documents and produces a structured data. This output is in a summarized form such as an excel sheet and table in a relational database.

Nowadays extraction is becoming a key in the big data industry.

As we know that the huge volume of data is getting generated out which most of the data is unstructured. The first key challenge is handling unstructured data. Now conversion of unstructured data to structured form based on some pattern so that the same can stored in RDBMS.

Apart from this in current days' data collection mechanism is also getting change. Earlier we collected data in batches like End-of-Day (EOD), but now business wants the data as soon as it is getting generated, i.e. in real time.

Regression

We can apply Machine learning to regression as well.

Assume that x= x1, x2, x3, … xn are the input variables and y is the outcome variable. In this case, we can use machine learning technology to produce the output (y) on the basis of the input variables (x). You can use a model to express the relationship between various parameters as below:

Y=g(x) where g is a function that depends on specific characteristics of the model. In regression, we can use the principle of machine learning to optimize the parameters. To cut the approximation error and calculate the closest possible outcome.

We can also use Machine learning for function optimization. We can choose to alter the inputs to get a better model. This gives a new and improved model to work with. This is known as response surface design.

CHAPTER 2
LITERATURE SURVEY

1. Spam Detection On Social Media Platforms

With the increased quality of online social platforms, spammers have come up with various techniques to lure users into accessing malicious links. This is done by generating spam on the comment section of various social media networks. In this paper, we've taken YouTube comments as the dataset and performed spam YouTube comment detection. The current methods to stop spammers include using tools such as Google Safe Browsing which help to detect and also block irrelevant spam on YouTube. Although these tools help in blocking harmful links, it fails to secure the users in real-time scenarios. Thus, many different approaches have been applied to form an environment that is spam free**.**

2. Combating Comment Spam with Machine Learning Approaches

In this paper, we propose and develop a comment spam detection mechanism that can be deployed as a browser plugin for inspecting the Document Object Model (DOM) of

the web page in question and remove comments with spam content. We examine most detection features in the literature along with proposing new features to build a comment spam classifier. In order to test the accuracy of our classifier, we manually label a new corpus of blogs comments. We encourage other researchers to build upon our work and we hope that our corpus will benefit the research community in this area.

3. Detecting Arabic YouTube Spam Using Data Mining Techniques

since YouTube became one of the sources of income, the number of users has increased significantly and the number of spammers who aim to spread viruses or to promote their videos and channels. These behaviors have led many YouTubers to close their channels or to disable the comments because YouTube does not have enough tools to prevent it. Filtering Arabic spam comments is a big challenge at all according to various dialects which hold a huge number of synonyms. In this work, we have classified these comments using different algorithms such as Decision Tree(DT), Support Vector Machine (SVM), Naive Bayes(NB), Random Forest, and k-Nearest Neighbor (k-NN).

4. SpamHD: Memory-Efficient Text Spam Detection using Brain-Inspired Hyperdimensional Computing

Brain-inspired hyperdimensional Computing (HDC) leverages the mathematical properties of high-dimensional vectors (hypervectors) which show remarkable agreement with how brain functions. Hypervectors (HVs) are high-dimensional (e.g., 10,000 dimensions), holographic, and (pseudo)random with independent and identically distributed (i.i.d) components. Recently, HDC has demonstrated promising capability in a wide range of applications such as robotics, bio-medical signal processing, and genome sequencing. Text spam detection is a classic natural language processing (NLP) task that is usually solved using machine learning methods associated with data preprocessing techniques such as tokenization.

5. Identification of spam comments using natural language processing techniques

The high popularity of modern web is partly due to the increase in the number of content sharing applications. The social tools provided by the content sharing applications allow online users to interact, to express their opinions and to read opinions from other users. However, spammers provide comments which are written intentionally to mislead users by redirecting them to web sites to increase their rating and to promote products less known on the market. Reading spam comments is a bad experience and a

waste of time for most of the online users but can also be harming and cause damage to the reader.

CHAPTER 3

SYSTEM ANALYSIS

3.1 EXISTING SYSTEM

- Research on detecting spam content and users focus on various fields. Many studies focused on spam on websites (e.g., portal sites and blogs). As YouTube gains popularity as a video sharing platform, spammers target it with low quality content or promotions. Since spammers that harm the YouTube community are increasing, detecting them becomes an interesting source to research. So, we divide the literature of detecting spam into two sections, spam on websites and spam on YouTube.

3.1.1 DISADVANTAGES OF EXISTING SYSTEM

3.2 PROPOSED SYSTEM

- Our proposed method is based on comparative research which is a representative study on YouTube spam comment detection. Our method applied six machine learning techniques to improve the performance of the Cascaded Ensemble Machine Learning Model aware YouTube Spam Comments Detection Scheme. These performed well in and were significant with 99.9% confidence. We propose an ensemble model combining them and evaluate the performance. The experimental environment used version 3.7.1 of Python and version 0.20.1 of the Cicely Library on Jupiter notebooks

3.2.1 ADVANTAGES OF PROPOSED SYSTEM

CHAPTER 4

SYSTEM REQUIREMENTS

The software requirements specification is produced at the culmination of the analysis task. The function and performance allocated to software as part of system engineering are refined by establishing a complete information description as functional representation of system behavior, an indication of performance requirements and design constraints, appropriate validation criteria.

4.1 HARDWARE AND SOFTWARE SPECIFICATION

The software requirements specification is produced at the culmination of the analysis task. The function and performance allocated to software as part of system engineering are refined by establishing a complete information description as functional representation of system behavior, an indication of performance requirements and design constraints, appropriate validation criteria.

4.2 Hardware Requirements

System : Pentium IV 2.4 GHz

Hard Disk : 40 GB

Floppy Drive : 1.44 Mb

Monitor : 15 VGA Colour

Mouse : Logitech

Ram : 512 Mb

CHAPTER 1

INTRODUCTION

.1 MACHINE LEARNING – OVERVIEW

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Following are the key machine learning tasks briefed later in this article:

- Feature selection
- Regression
- Classification
- Clustering
- Multivariate querying
- Density estimation
- Dimension reduction
- Testing and matching

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- Ridge (L2) regularization

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- Gaussian process regression *(Higher accuracy)*
- Regression trees
- Linear regression
- Support vector regression
- LASSO

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- Artificial neural networks (ANN) *(Higher accuracy)*
- Support vector machine (SVM) *(Higher accuracy)*
- Random forests *(Higher accuracy)*
- Decision trees
- Boosted trees
- Logistic regression
- naive Bayes

DEEP LEARNING

Clustering: Clustering tasks are all about finding natural groupings of data and a label associated with each of these groupings (clusters). Some of the common example includes customer segmentation, product features identification for product roadmap. Some of the following are common ML methods:

- Mean-shift (*Higher accuracy)*
- Hierarchical clustering
- K-means
- Topic models

Multivariate querying: Multivariate querying is about querying or finding similar objects. Some of the following ML methods could be used for such problems:

- Nearest neighbors
- Range search
- Farthest neighbors

Density estimation: Density estimation problems are related with finding likelihood or frequency of objects. In probability and statistics, density estimation is the construction of an estimate, based on observed data, of an unobservable underlying probability density function. Some of the following ML methods could be used for solving density estimation tasks:

- Kernel density estimation *(Higher accuracy)*
- Mixture of Gaussians
- Density estimation tree

Dimension reduction: Dimension reduction is the process of reducing the number of random variables under consideration, and can be divided into feature selection and feature extraction. Following are some of ML methods that could be used for dimension reduction:

- Manifold learning/KPCA *(Higher accuracy)*
- Principal component analysis
- Independent component analysis
- Gaussian graphical models

- Non-negative matrix factorization
- Compressed sensing.

Testing and matching: Testing and matching tasks relates to comparing data sets. Following are some of the methods that could be used for such kind of problems:

- Minimum spanning tree
- Bipartite cross-matching
- N-point correlation

.3 MACHINE LEARNING SYSTEM CLASSIFICATION

Although supervised and unsupervised learning are two of the most widely accepted machine learning methods by businesses today, there are various other machine learning techniques. Following is an overview of some of the most accepted ML methods

SUPERVISED LEARNING

These algorithms are trained using labeled examples, in different scenarios, as an input where the desired outcome is already known. An equipment, for instance, could have data points such as "F" and "R" where "F" represents "failed" and "R" represents "runs".

A learning algorithm will receive a set of input instructions along with the corresponding accurate outcomes. The learning algorithm will then compare the actual outcome with the accurate outcome and flag an error, if there is any discrepancy. Using different methods, such as regression, classification, gradient boosting, and prediction, supervised learning uses different patterns to proactively predict the values of a label on extra unlabeled data. This method is commonly used in areas where historical data is used to predict events that are likely to occur in the future.

UNSUPERVISED LEARNING

This method of ML finds its application in areas were data has no historical labels. Here, the system will not be provided with the "right answer" and the algorithm should identify what is being shown. The main aim here is to analyze the data and identify a pattern and structure within the available data set. Transactional data serves as a good source of data set for unsupervised learning.

For instance, this type of learning identifies customer segments with similar attributes and then lets the business to treat them similarly in marketing campaigns. Similarly, it can also identify attributes that differentiate customer segments from one another. Either ways, it is about identifying a similar structure in the available data set. Besides, these algorithms can also identify outliers in the available data sets.

Some of the widely used techniques of unsupervised learning are

- k-means clustering
- self-organizing maps
- value decomposition
- mapping of nearest neighbor.

SEMI-SUPERVISED LEARNING

This kind of learning is used and applied to the same kind of scenarios where supervised learning is applicable. However, one must note that this technique uses both unlabeled and labeled data for training. Ideally, a small set of labeled data, along with a large volume of unlabeled data is used, as it takes less time, money and efforts to acquire unlabeled data. This type of machine learning is often used with methods, such as regression, classification and prediction.

REINFORCEMENT LEARNING

This is mainly used in navigation, robotics and gaming. Actions that yield the best rewards are identified by algorithms that use trial and error methods. There are three major components in reinforcement learning, namely, the agent, the actions and the

environment.

1.4 MACHINE LEARNING APPLICATIONS

As we move forward into the digital age, one of the modern innovations we've seen is the creation of Machine Learning. This incredible form of artificial intelligence is already being used in various industries and professions. For Example, Image and Speech Recognition, Medical Diagnosis, Prediction, Classification, Learning Associations, Statistical Arbitrage, Extraction, Regression. Today we're looking at all these Machine Learning Applications in today's modern world.

Image Recognition

It is one of the most common machine learning applications. There are many situations where you can classify the object as a digital image. For digital images, the measurements describe the outputs of each pixel in the image.

In the case of a black and white image, the intensity of each pixel serves as one measurement. So if a black and white image has N*N pixels, the total number of pixels and hence measurement is N2.

In the colored image, each pixel considered as providing 3 measurements of the intensities of 3 main color components i.e. RGB. So N*N colored image there are 3 N2 measurements.

For face detection – The categories might be face versus no face present. There might be a separate category for each person in a database of several individuals.
For character recognition – We can segment a piece of writing into smaller images, each containing a single character. The categories might consist of the 26 letters of the English alphabet, the 10 digits, and some special characters.

Speech Recognition

Speech recognition (SR) is the translation of spoken words into text. It is also known

as “automatic speech recognition” (ASR), “computer speech recognition”, or “speech to text” (STT).

In speech recognition, a software application recognizes spoken words. The measurements in this Machine Learning application might be a set of numbers that represent the speech signal. We can segment the signal into portions that contain distinct words or phonemes. In each segment, we can represent the speech signal by the intensities or energy in different time-frequency bands.
Although the details of signal representation are outside the scope of this program, we can represent the signal by a set of real values.

Speech recognition, Machine Learning applications include voice user interfaces. Voice user interfaces are such as voice dialing, call routing, demotic appliance control. It can also use as simple data entry, preparation of structured documents, speech-to-text processing, and plane.

Medical Diagnosis

ML provides methods, techniques, and tools that can help in solving diagnostic and prognostic problems in a variety of medical domains. It is being used for the analysis of the importance of clinical parameters and of their combinations for prognosis, e.g. prediction of disease progression, for the extraction of medical knowledge for outcomes research, for therapy planning and support, and for overall patient management. ML is also being used for data analysis, such as detection of regularities in the data by appropriately dealing with imperfect data, interpretation of continuous data used in the Intensive Care Unit, and for intelligent alarming resulting in effective and efficient monitoring.

It is argued that the successful implementation of ML methods can help the integration of computer-based systems in the healthcare environment providing opportunities to facilitate and enhance the work of medical experts and ultimately to improve the efficiency and quality of medical care.

In medical diagnosis, the main interest is in establishing the existence of a disease followed by its accurate identification. There is a separate category for each disease under consideration and one category for cases where no disease is present. Here, machine learning improves the accuracy of medical diagnosis by analyzing data of patients.

The measurements in this Machine Learning applications are typically the results of certain medical tests (example blood pressure, temperature and various blood tests) or medical diagnostics, presence/absence/intensity of various symptoms and basic physical information about the patient (age, sex, weight etc.). On the basis of the results of these measurements, the doctors narrow down on the disease inflicting the patient.

Statistical Arbitrage

In finance, statistical arbitrage refers to automated trading strategies that are typical of a short-term and involve a large number of securities. In such strategies, the user tries to implement a trading algorithm for a set of securities on the basis of quantities such as historical correlations and general economic variables. These measurements can be cast as a classification or estimation problem. The basic assumption is that prices will move towards a historical average.
We apply machine learning methods to obtain an index arbitrage strategy. In particular, we employ linear regression and support vector regression (SVR) onto the prices of an exchange-traded fund and a stream of stocks. By using principal component analysis (PCA) in reducing the dimension of feature space, we observe the benefit and note the issues in the application of SVR. To generate trading signals, we model the residuals from the previous regression as a mean reverting process.

In the case of classification, the categories might be sold, buy or do nothing for each security. I the case of estimation one might try to predict the expected return of each security over a future time horizon. In this case, one typically needs to use the estimates of the expected return to make a trading decision (buy, sell, etc.)

Learning Associations

Learning association is the process of developing insights into various associations between products. A good example is how seemingly unrelated products may reveal an association to one another. When analyzed in relation to buying behaviors of customers.

One application of machine learning- Often studying the association between the products people buy, which is also known as basket analysis. If a buyer buys 'X', would he or she force to buy 'Y' because of a relationship that can identify between them? This leads to the relationship that exists between fish and chips etc.

when new products launch in the market a Knowing these relationships it develops a new relationship. Knowing these relationships could help in suggesting the associated product to the customer. For a higher likelihood of the customer buying it, it can also help in bundling products for a better package.

This learning of associations between products by a machine is learning associations. Once we found an association by examining a large amount of sales data, Big Data analysts. It can develop a rule to derive a probability test in learning a conditional probability.

Classification

Classification is a process of placing each individual from the population under study in many classes. This is identified as independent variables.

Classification helps analysts to use measurements of an object to identify the category to which that object belongs. To establish an efficient rule, analysts use data. Data consists of many examples of objects with their correct classification.

For example, before a bank decides to disburse a loan, it assesses customers on their ability to repay the loan. By considering factors such as customer's earning, age, savings and financial history we can do it. This information is taken from the past data of the loan. Hence, Seeker uses to create a relationship between customer attributes and related risks.

Prediction

Consider the example of a bank computing the probability of any of loan applicants faulting the loan repayment. To compute the probability of the fault, the system will first need to classify the available data in certain groups. It is described by a set of rules prescribed by the analysts.

Once we do the classification, as per need we can compute the probability. These probability computations can compute across all sectors for varied purposes

The current prediction is one of the hottest machine learning algorithms. Let's take an example of retail, earlier we were able to get insights like sales report last month / year / 5-years / Diwali / Christmas.

Extraction

Information Extraction (IE) is another application of machine learning. It is the process of extracting structured information from unstructured data. For example, web pages, articles, blogs, business reports, and e-mails. The relational database maintains the output produced by the information extraction.

The process of extraction takes input as a set of documents and produces a structured data. This output is in a summarized form such as an excel sheet and table in a relational database.

Nowadays extraction is becoming a key in the big data industry.

As we know that the huge volume of data is getting generated out which most of the data is unstructured. The first key challenge is handling unstructured data. Now conversion of unstructured data to structured form based on some pattern so that the same can stored in RDBMS.

Apart from this in current days' data collection mechanism is also getting change. Earlier we collected data in batches like End-of-Day (EOD), but now business wants the data as soon as it is getting generated, i.e. in real time.

Regression

We can apply Machine learning to regression as well.

Assume that x= x1, x2, x3, … xn are the input variables and y is the outcome variable. In this case, we can use machine learning technology to produce the output (y) on the basis of the input variables (x). You can use a model to express the relationship between various parameters as below:

Y=g(x) where g is a function that depends on specific characteristics of the model. In regression, we can use the principle of machine learning to optimize the parameters. To cut the approximation error and calculate the closest possible outcome.

We can also use Machine learning for function optimization. We can choose to alter the inputs to get a better model. This gives a new and improved model to work with. This is known as response surface design.

CHAPTER 2
LITERATURE SURVEY

1. Spam Detection On Social Media Platforms

With the increased quality of online social platforms, spammers have come up with various techniques to lure users into accessing malicious links. This is done by generating spam on the comment section of various social media networks. In this paper, we've taken YouTube comments as the dataset and performed spam YouTube comment detection. The current methods to stop spammers include using tools such as Google Safe Browsing which help to detect and also block irrelevant spam on YouTube. Although these tools help in blocking harmful links, it fails to secure the users in real-time scenarios. Thus, many different approaches have been applied to form an environment that is spam free**.**

2. Combating Comment Spam with Machine Learning Approaches

In this paper, we propose and develop a comment spam detection mechanism that can be deployed as a browser plugin for inspecting the Document Object Model (DOM) of the web page in question and remove comments with spam content. We examine most

detection features in the literature along with proposing new features to build a comment spam classifier. In order to test the accuracy of our classifier, we manually label a new corpus of blogs comments. We encourage other researchers to build upon our work and we hope that our corpus will benefit the research community in this area.

3. Detecting Arabic YouTube Spam Using Data Mining Techniques

since YouTube became one of the sources of income, the number of users has increased significantly and the number of spammers who aim to spread viruses or to promote their videos and channels. These behaviors have led many YouTubers to close their channels or to disable the comments because YouTube does not have enough tools to prevent it. Filtering Arabic spam comments is a big challenge at all according to various dialects which hold a huge number of synonyms. In this work, we have classified these comments using different algorithms such as Decision Tree(DT), Support Vector Machine (SVM), Naive Bayes(NB), Random Forest, and k-Nearest Neighbor (k-NN).

4. SpamHD: Memory-Efficient Text Spam Detection using Brain-Inspired Hyperdimensional Computing

Brain-inspired hyperdimensional Computing (HDC) leverages the mathematical properties of high-dimensional vectors (hypervectors) which show remarkable agreement with how brain functions. Hypervectors (HVs) are high-dimensional (e.g., 10,000 dimensions), holographic, and (pseudo)random with independent and identically distributed (i.i.d) components. Recently, HDC has demonstrated promising capability in a wide range of applications such as robotics, bio-medical signal processing, and genome sequencing. Text spam detection is a classic natural language processing (NLP) task that is usually solved using machine learning methods associated with data preprocessing techniques such as tokenization.

5. Identification of spam comments using natural language processing techniques

The high popularity of modern web is partly due to the increase in the number of content sharing applications. The social tools provided by the content sharing applications allow online users to interact, to express their opinions and to read opinions from other users. However, spammers provide comments which are written intentionally to mislead users by redirecting them to web sites to increase their rating and to promote products less known on the market. Reading spam comments is a bad experience and a waste of time for most of the online users but can also be harming and cause damage to

the reader.

CHAPTER 3

SYSTEM ANALYSIS

3.1 EXISTING SYSTEM

- Research on detecting spam content and users focus on various fields. Many studies focused on spam on websites (e.g., portal sites and blogs). As YouTube gains popularity as a video sharing platform, spammers target it with low quality content or promotions. Since spammers that harm the YouTube community are increasing, detecting them becomes an interesting source to research. So, we divide the literature of detecting spam into two sections, spam on websites and spam on YouTube.

3.1.1 DISADVANTAGES OF EXISTING SYSTEM

3.2 PROPOSED SYSTEM

- Our proposed method is based on comparative research which is a representative study on YouTube spam comment detection. Our method applied six machine learning techniques to improve the performance of the Cascaded Ensemble Machine Learning Model aware YouTube Spam Comments Detection Scheme. These performed well in and were significant with 99.9% confidence. We propose an ensemble model combining them and evaluate the performance. The experimental environment used version 3.7.1 of Python and version 0.20.1 of the Cicely Library on Jupiter notebooks

3.2.1 ADVANTAGES OF PROPOSED SYSTEM

CHAPTER 4

SYSTEM REQUIREMENTS

The software requirements specification is produced at the culmination of the analysis task. The function and performance allocated to software as part of system engineering are refined by establishing a complete information description as functional representation of system behavior, an indication of performance requirements and design constraints, appropriate validation criteria.

4.1 HARDWARE AND SOFTWARE SPECIFICATION

The software requirements specification is produced at the culmination of the analysis task. The function and performance allocated to software as part of system engineering are refined by establishing a complete information description as functional representation of system behavior, an indication of performance requirements and design constraints, appropriate validation criteria.

4.2 Hardware Requirements

System : Pentium IV 2.4 GHz

Hard Disk : 40 GB

Floppy Drive : 1.44 Mb

Monitor : 15 VGA Colour

Mouse : Logitech

Ram : 512 Mb

4.3 Software Prerequisites

Windows 10 as the operating system

Anaconda Navigator is the IDE.

Python as the coding language

CHAPTER 5

Systems Design

System architecture (5.1)

Design is a multi-step process that concentrates on the interface between modules, procedural details, programme architecture, and data structures. Before any coding is done, the design process also decodes the requirements into a presentation of software

that can be accessed for quality. The design of computer programmes changes regularly as new techniques, better analysis techniques, and border understanding emerge. The revolution in software proposal is still at relatively early stages.

The depth, flexibility, and quantitative nature typically associated with more traditional engineering disciplines are not present in software design methodology. However, there are software design approaches, design quality standards, and design notation that can be used.

5.2 ARCHITECTURE DIAGRAM:

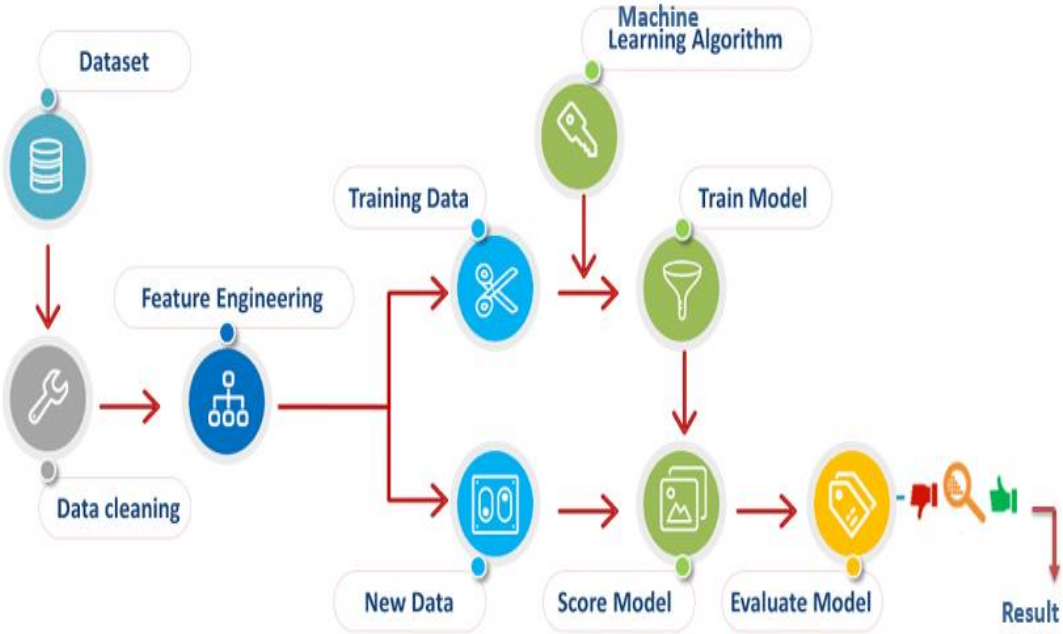


Fig 5.2 System Architecture

5.3 FLOW DIAGRAM

Flow diagram is a collective term for a diagram representing a flow or set of dynamic relationships in a system. The term flow diagram is also used as a synonym for flowchart, and sometimes as a counterpart of the flowchart.

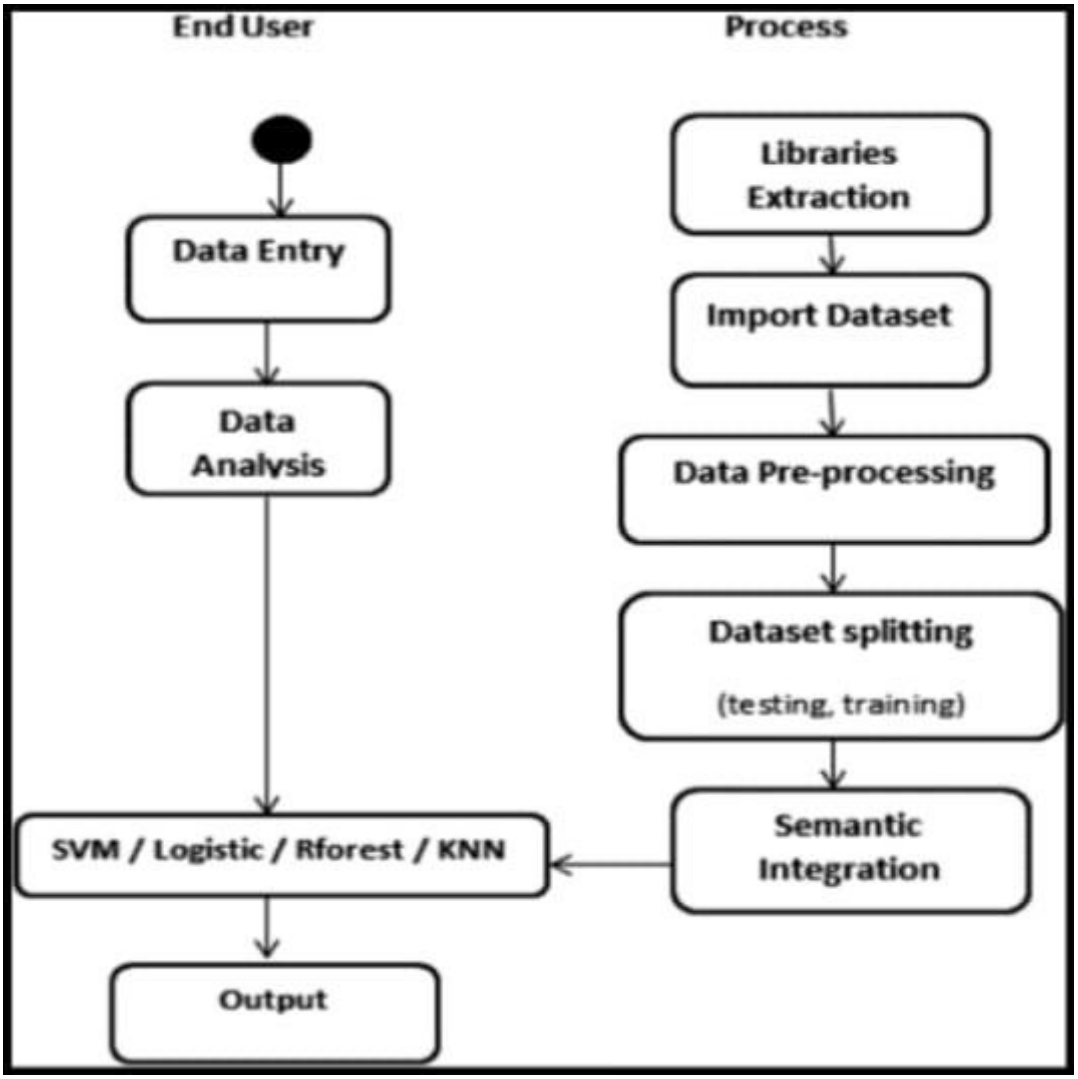


Fig 5.3 Flow Diagram

CHAPTER 6

IMPLEMENTATION OF SYSTEM

Introduction: 6.1

Following the selection of the profile, the appropriate qualities (i.e., features) are chosen on which to base the classification method. As new training data is fed into the classifier on a regular basis,4. the classifier is trained. It is determined by the classifier if a profile is false or real.

System modules (6.2)

Getting a Dataset

Pre-processing

cleansing of data

transformation of data

choosing data

data entry

Algorithm

Simple Bayes

Result

6.3 DESCRIPTION OF MODULE:

Getting a Dataset

One of the most crucial steps in creating a machine learning model is data collection.

We gather the particular dataset in accordance with the demands from the internet.

Additionally, the dataset includes some undesired data. Therefore, we must first pre-process the data to create the ideal data set for the algorithm.

Pre-processing of data

Data cleansing, data transformation, and data selection are all forms of data pre-processing.

Fill in missing numbers, smooth noisy data, locate or get rid of outliers, and deal with inconsistencies when cleaning data.

Data transformation techniques that enhance the quality of the data include smoothing, aggregating, generalisation, and transformation.

Data selection covers a few techniques or features that let us choose the pertinent data for our system.

6.4 FORMULA

Dataset values were translated into array values, which the algorithm would use to determine correctness.

Analyse the data using the algorithm after choosing it based on accuracy.

Algorithm for Naive Bayes Classifier

The Nave Bayes algorithm is a supervised learning method for classification issues that is based on the Bayes theorem.

It is mostly employed in text categorization with a large training set.

The Naive Bayes Classifier is one of the most straightforward and efficient classification algorithms available today. It aids in the development of quick machine learning models capable of making accurate predictions.

Being a probabilistic classifier, it makes predictions based on the likelihood that an object will occur.

Spam filtration, Sentimental analysis, and article classification are a few examples of Naive Bayes algorithms that are frequently used.

The Bayes Theorem

The Bayes theorem, commonly referred to as Bayes' Rule or Bayes' law, is used to calculate the likelihood of a hypothesis given some prior information. The conditional probability determines this.

The Bayes theorem's formula is as follows:

Where,

The posterior probability, or P(A|B), measures the likelihood that a given hypothesis (A) will really occur.

P(B|A) stands for Likelihood Probability, which measures how likely it is based on the evidence at hand that a given hypothesis is correct.

Priority probability, or P(A), is the likelihood of a theory before seeing the evidence.

The probability of evidence is marginal probability, or P(B).

Naive Bayes' Classifier Operation:

The following example will allow you to understand how the Naive Bayes' Classifier functions:

Assume that "Play" is the equivalent target variable and that we have a dataset of weather conditions. Therefore, utilising this dataset, we must determine whether or not to play on a specific day based on the weather. In order to solve this issue, we must take the following actions:

Create frequency tables from the provided dataset.

Create a likelihood table by calculating the odds of the given attributes.

Now, determine the posterior probability using the Bayes theorem.

Question: Should the Player play if the weather is sunny?

Solution: To begin, think about the dataset shown below:

Naive Bayes Classifier benefits include:

One of the quick and simple machine learning techniques to predict a class of datasets is naive Bayes.

Both binary and multi-class classifications can be done with it.

In comparison to other algorithms, it performs well in multi-class predictions.

It is the most often used solution for text classification issues.

Naive Bayes Classifier drawbacks:

Naive Bayes cannot discover the relationship between features because it presumes that all features are either independent or unrelated.

Uses for the Nave Bayes Classifier

For credit scoring, it is employed.

It is applied to the classification of medical data.

Due of its eagerness to learn, the Nave Bayes Classifier can make predictions in real time.

It is utilised in text classification processes like Sentiment analysis and spam screening.

RESULT:

Based on that dataset, we can forecast the outcome using our random forest method.

Here, we may also learn how accurate the prediction was.

Finding spam websites will be made easier by it.

CHAPTER 7

SOFTWARE ENVIRONMENT

PYTHON TECHNOLOGY 7.1

Python definition

Python is a dynamic, high-level, general-purpose, and interpreted programming language. It allows the development of programmes using an object-oriented approach. It offers a large number of high-level data structures and is straightforward and simple to learn.

Python is a programming language that is appealing for application development since it is simple to learn but strong and flexible.

Python is the best language for scripting and quick application development because of its syntax, dynamic typing, and nature as an interpreted language.

Python supports a variety of programming patterns, including imperative, functional, and object-oriented programming patterns.

Python is not designed to be used for a certain task, like web programming. Because it can be utilised with online, enterprise, 3D CAD, etc., it is known as a multipurpose programming language.

Since the variables are dynamically typed, we can simply write a=10 to assign an integer value to an integer variable without using data types to specify them.

Because there is no compilation stage involved in Python development and the edit-test-debug cycle is relatively quick, Python makes development and debugging quick.

Python 3 versus Python 2.

When a new version of a programming language is released, the majority of them maintain the features and syntax of the previous version, making it simpler for projects to convert to the newer version. The two Python versions, Python 2 and Python 3, are, however, significantly dissimilar from one another.

The following is a list of the variations between Python 2 and Python 3:

1. To print a string on the console, Python 2 employs the print statement and the print "something" command. To print something on the terminal, Python 3 employs the print function, which is used like print("something").
2. The raw_input() method in Python 2 is used to take user input. It provides the user-typed value's string representation in return. We must utilise Python's int() method to convert it to an integer. However, Python 3 employs the input() function, which does automatic interpretation of the user's input. However, by using primitive operations (such int(), str(), etc.), we can cast this value to any type.

3. The implicit string type in Python 2 is ASCII, whereas the implicit string type in Python 3 is Unicode.

4. The Python 2 xrange() method is absent from Python 3. The xrange() function is a range() variation that produces an object that functions similarly to a Java iterator. The function range() returns a list, for instance, range(0,3) contains 0, 1, and 2.

5. Additionally, Python 3's handling of exceptions has undergone a minor adjustment. It describes a keyword as one that must be utilised. It will be covered in the Python programming tutorial's section on managing exceptions.

Python History

Guido van Rossum created Python in 1991 at CWI in the Netherlands. The ABC

programming language served as a model for the Python programming language, so we could argue that ABC was Python's forerunner.

The name Python was chosen for a reason as well. The then-famous BBC comedy programme "Monty Python's Flying Circus" was a favourite of Guido van Rossum's. So he made the decision to give his newly developed programming language the name Python.

Python has a sizable international community and quickly updates its versions.

Why learn Python?

The programmer can take advantage of numerous valuable features in Python. Due to these qualities, it is the most extensively spoken and used language. The essential Python features are described below.

simple to learn and use

Communicative Language

Translated Language

Why learn Python?

The programmer can take advantage of numerous valuable features in Python. Due to these qualities, it is the most extensively spoken and used language. The essential Python features are described below.

simple to learn and use

Communicative Language

Translated Language

Language That Is Object-Oriented

Open Source Programming

Language That Is Object-Oriented

Open Source Programming

Study the Common Library

Support for GUI Programming

Integrated

Embeddable

Allocating Memory Dynamically

Libraries and frameworks in a wide variety.

Where is Python used?

Python is a widely used, general-purpose programming language that is employed in practically all technical fields. Below is a list of the various applications for Python.

Science of Data

Dating back

Using Desktop Programmes

Applications based on consoles

Applications for Mobile

Application Development

Machine intelligence

Web-based programmes

Business Applications

3-D CAD software

Learning Machines

Applications for computer vision or image processing.

Recognition of speech

Standard Python Syntax

Curly brackets and semicolons are not permitted in Python code. It sounds a lot like English. Python, however, makes use of indentation to identify a block of code. When a statement needs more space before it can be read, it is indented.

For example -

```
def func():
        statement 1
        statement 2
        ……————————————
        ……————————————
          statement N
```

The statements in the forementioned example that are level right belong to the function. Typically, indentation is defined by four whitespaces.

First Python Program

Python offers the ability to run code in less lines than other programming languages.For instance, if we were to display the Java "Hello World" programme, it would require three lines to do it.

```
1. public class HelloWorld {
2.  public static void main(String[] args){
3. // Prints "Hello, World" to the terminal window.
4.   System.out.println("Hello World");
5.  }
6.  }
```

On the other hand, we can do this using one statement in Python.

```
1. print("Hello World")
```

Both programmes will display the same output, however Python just requires a single statement without the need of curly brackets or a semicolon.

Popular Frameworks and Libraries for Python

In many different domains, including machine learning, artificial intelligence, web applications, etc., Python has a large variety of libraries and frameworks that are often utilised. Here is a list of some well-known Python frameworks and libraries.

Server-side web development using Django Flask, Pyramid, and CherryPy

Applications that use GUIs include Tk, PyGTK, PyQt, PyJs, etc.

TensorFlow, PyTorch, Scikit-Learn, Matplotlib, Scipy, and other machine learning tools.

Mathematics: Pandas, Numpy, etc.

Python print() Function

The print() function outputs the specified object to either a text stream file or the standard output device (screen).

In contrast to other programming languages, Python's print() method is the most distinctive and flexible.

The syntax of **print()** function is given below.

1. print(*objects, sep=' ', end='\n', file=sys.stdout, flush=False)

Let's go over each of its components one by one.

objects - A statement intended to be printed is all that an object is. Multiple statements are indicated by the symbol *.

sep - The print values are separated by the sep parameter. "is the default value.

end - The statement's final word is printed.

A file must have a write(string) method and be an object.

If flush is true, the stream or file is forcibly flushed. Its value is false by default.

Let's analyse the next illustration.

Operators in Python

The symbols known as operators are used to manipulate Python objects in various ways. The most crucial tools for interacting with Python data types are Python operators. Additionally, Python offers bitwise and identify membership operators. In the tutorial that follows, we will study all of these operators with the appropriate examples.

Operators in Python

Conditional Statements in Python

We can use conditional statements to run a specific block in response to a specific circumstance. This article will teach you how to execute a different set of statements using a conditional expression. The if and else keywords in Python are used to create logical situations. Elif is also employed as a conditional statement.

if..else statement in Python

Loops in Python

The direction of the programme may need to be changed occasionally. It could be necessary to execute a particular code multiple times. Programming languages offer a variety of loop types that can repeat a piece of code repeatedly for this purpose. For a detailed understanding of the assertions, take a look at the tutorial below.

Loops in Python

Python For Loop

Python While Loop

Python Data Structures

When referring to a group of data that can be stored together or in an organised form, we use the term "data structure." Data structures including list, tuple, dictionary, and set are included in Python. We can use data structures to carry out complicated tasks.

Python List

The sorted collection of things is stored in a Python list. A list can be used to hold a list of elements in order. Python lists are changeable, which means they can be changed after they've been created. List items are delimited by a square bracket ([]) and separated by commas.

Python Tuple

The sequence of immutable Python objects is stored in a Python Tuple. The tuple is similar to lists in that the values of the items stored in the list can be modified, but the values of the items stored in the tuple cannot be changed since the tuple is immutable.
Python String

A string in Python is a collection of characters. It consists of a group of characters enclosed in single, double, or triple quotations. It is also possible to define it as a group of Unicode characters.

Dictionaries

The most effective data structure for storing a lot of data is the Python dictionary. The information is kept in a key-value pair format. Each value is stored alongside its matching key.

Unique keys are required, and values can be of any type, including integer, list, tuple, etc.

It is a changeable type, thus we can change its assignment once it is made.

Sets in Python

In Python, a set is a group of unsorted elements. The set's components must all be distinct and unchangeable. Because sets are mutable, we can change them at any time during the course of the programme.

Functional Programming in Python

The definitions of key functional programming tools like lambda and recursive functions are provided in this portion of the Python course. These processes are particularly effective at completing difficult tasks. We define some key operations including reduce, map, and filter.The functools module of Python comes with a number of functional programming tools.

- Python Function
- Python map() Function
- Python filter() Function
- Python reduce() Function
- Python functool Module
- Python Lambda Function

Python File I/O

Data is stored as files on a computer disc. We describe Python's built-in file object in this tutorial. Using a Python script, we can open a file and carry out a number of activities, including writing, reading, and appending. A file can be opened in a number of different ways.

Python File I/O

Python Modules

The programme files that contain Python code or functions are known as modules. User-define modules and built-in modules are the two different types of modules available in Python. A user-defined module, or, alternatively, Python code saved with the.py extension, is regarded as such.

Python has prepackaged modules called built-in modules. We must import the modules into our running programme in order to use their features.

Python Modules

Python Exceptions

An uncommon circumstance in a programme that interrupts the flow of the programme is known as an exception.

When an exception occurs, the programme halts execution and does not run any additional code. Run-time failures that a Python script cannot manage are thus an exception. A Python object that describes an error is called an exception.

Python Exceptions

CSV in Python

The term "comma separated values" (or "csv") refers to a straightforward file format that uses certain formatting to arrange tabular data. It provides a standard format for data interchange and stores tabular data, such as from a spreadsheet or database, in plain text. The rows and columns of data in a csv file that opens in an excel sheet determine the format that should be used. For a thorough explanation of the CSV module, visit the lesson below.

Python Read CSV File

Python Write CSV File

Python Sending Mail

Using the Python script, we are able to send and receive email. The standard library modules for Python are helpful for managing different protocols like PoP3 and IMAP. We'll discover how to use a Python script to send an email using the well-known email service SMTP.

Python Sending Emails

Python Magic Methods

The unique method that adds "magic" to a class is referred to as a "magic method" in Python. It begins and finishes with double underscores, such as in the words "init" or "str."

Many magic methods are defined by the built-in classes. You can use the dir() function to determine how many magic methods a class has inherited. The method name contains two prefixes and an underscore at the end.

Python Magic Methods

Python Oops Concepts

Python treats everything as an object, including float values, functions, classes, and none. Python also supports all other oriented concepts. Here is a quick overview of Python's oops principles.

- **Classes and Objects -** Python classes are the blueprint of the object. An object is a collection of data and method that act on the data.

- Inheritance - An inheritance is a technique where one class inherits the properties of other classes.
- Constructor - Python provides a special method __init__() which is known as a constructor. This method is automatically called when an object is instantiated.
- Data Member - A variable that holds data associated with a class and its objects.

To read the oops concept in detail, visit the following resources.

- Python Oops Concepts
- Python Object and classes
- Python Constructor
- Python Inheritance
- Python Polymorphism

Python Advance Topics

Python has a wealth of sophisticated and practical features that assist programmers in handling challenging tasks. The concepts below are listed.

Python Iterator

A simple object that can be iterated upon is an iterator. One object at a time is returned. The two special methods, __iter__() and __next__(), can be used to implement it.

Visit our Python Iterators tutorial to learn more about iterators.

Python Generators

It is simplest to create Iterators using Generators. Visit our Python Generators tutorial to find out more.

Python Decorators

These are used to alter how the function behaves. Decorators provide you the freedom to increase the functionality of a wrapped function without permanently changing it by wrapping another function.

Python Database Connections

Python supports a variety of databases. Visit the links listed below to view the complete tutorial. Standard sets of functionality are acknowledged by the Python DBI-API to be present in the database connectivity modules for corresponding RDBMS solutions. We use Python DBI-API to demonstrate every significant database connectivity.

Python MySQL

Environment Setup

Database Connection

Creating New Database

Creating Tables

Insert Operation

Read Operation

Update Operation

Join Operation

Performing Transactions

Python MongoDB

Python MongoDB

Python SQLite

Python SQLite

Python CGI

Python CGI stands for "Common Gateway Interface", which is used to define how to exchange information between the webserver and a custom Python scripts. The Common Gateway Interface is a standard for external gateway programs to interface with the server, such as HTTP Servers. To learn more about Python CGI, visit the following tutorial.

- Python CGI

7.2 PYTHON PLATFORM

Apart from Windows, Linux and Marcos, CPython implementation runs on 21 different platforms. Iron Python is a .NET framework based Python implementation and it is capable of running in both Windows, Linux and in other environments where .NET framework is available.

7.3 PYTHON LIABRARY

Machine Learning, as the name suggests, is the science of programming a computer by which they are able to learn from different kinds of data. A more general definition given by Arthur Samuel is – "Machine Learning is the field of study that gives computers the ability to learn without being explicitly programmed." In the older days, people used to perform Machine Learning tasks by manually coding all the algorithms and mathematical and statistical formula. This made the process time consuming, tedious and inefficient. But in the modern days, it is become very much easy and efficient compared to the olden days by various python libraries, frameworks, and modules. Today, Python is one of the most popular programming languages for this task and it has replaced many languages in the industry, one of the reason is its vast collection of libraries.

Python libraries that used in Machine Learning are:

- Numpy
- Scipy
- Scikit-learn
- Theano
- Tensor Flow
- Keras
- PyTorch
- Pandas
- Matplotlib

NumPy

NumPy is a well-known Python toolkit for processing huge multidimensional arrays and matrices with the aid of a variety of sophisticated mathematical operations. For basic scientific computations in machine learning, it is highly helpful. Its capabilities in linear algebra, the Fourier transform, and random numbers are particularly advantageous. High-end libraries like Tensor Flow manipulate tensors internally using NumPy.

SciPy

Because it includes several modules for optimisation, linear algebra, integration, and statistics, SciPy is a very well-liked library among those interested in machine learning. The SciPy stack and SciPy library are distinct from one another. One of the essential components of the SciPy stack is SciPy. SciPy is a fantastic tool for manipulating images.

Skikit

Skikit-learn is one of the most popular ML libraries for classical ML algorithms. It is built on top of two basic Python libraries, viz., NumPy and SciPy. Scikit-learn supports most of the supervised and unsupervised learning algorithms. Scikit-learn can also be used for data-mining and data-analysis, which makes it a great tool who is starting out with ML.

Theano

We are all aware that mathematics and statistics are the foundation of machine learning. Thea no is a well-known Python library that is employed to effectively design, assess, and optimise mathematical expressions using multi-dimensional arrays. It is accomplished by optimising CPU and GPU usage. It is widely used to identify and diagnose various kinds of issues during unit testing and self-verification. Thea is a very

strong library that has long been used in complex, computationally demanding scientific projects, but it is straightforward and usable enough for individuals to use it for their own projects.

Tensor Flow

Tensor Flow is a very popular open-source library for high performance numerical computation developed by the Google Brain team in Google. As the name suggests, Tensor flow is a framework that involves defining and running computations involving tensors. It can train and run deep neural networks that can be used to develop several AI applications. Tensor Flow is widely used in the field of deep learning research and application.

Keras

Keras is a very popular Machine Learning library for Python. It is a high-level neural networks API capable of running on top of Tensor Flow, CNTK, or Thea no. It can run seamlessly on both CPU and GPU. Keras makes it really for ML beginners to build and design a Neural Network. One of the best thing about Keras is that it allows for easy and fast prototyping.

PyTorch

PyTorch is a popular open-source Machine Learning library for Python based on Torch, which is an open-source Machine Learning library which is implemented in C with a wrapper in Lau. It has an extensive choice of tools and libraries that supports on Computer Vision, Natural Language Processing(NLP) and many more ML programs. It allows developers to perform computations on Tensors with GPU acceleration and also helps in creating computational graphs.

Pandas

Pandas is a popular Python library for data analysis. It is not directly related to Machine Learning. As we know that the dataset must be prepared before training. In this case, Pandas comes handy as it was developed specifically for data extraction and preparation. It provides high-level data structures and wide variety tools for data analysis. It provides many inbuilt methods for groping, combining and filtering data.

Matpoltlib

Matpoltlib is a very popular Python library for data visualization. Like Pandas, it is not directly related to Machine Learning. It particularly comes in handy when a programmer wants to visualize the patterns in the data. It is a 2D plotting library used for creating 2D graphs and plots. A module named pyplot makes it easy for programmers for plotting as it provides features to control line styles, font properties, formatting axes, etc. It provides various kinds of graphs and plots for data visualization, viz., histogram, error charts, bar chats, etc,

CHAPTER 8
TESTING

The goal of testing is to find flaws. A work invention is tested in an effort to find any flaws or weaknesses that may exist. It is the process of training software with the intention of guaranteeing that the Software system fulfils its needs and user opportunities and does not fail in an appalling way. It gives a way to test the functionality of workings, sub-assemblies, meetings, and/or a finished product. Different test types exist. Every test type targets a certain testing circumstance.

8.1 TESTING OBJECTIVES

- All field entries must work properly.
- Pages must be started from the recognized link.
- The access screen, messages and replies must not be delayed.
- Features to be verified.
- Verify that the accesses are of the correct setup

- No replacement passes should be allowed.
- All links must take the user to the right page.

8.2 TESTING METHODOLOGIES

8.2.1 UNIT TESTING

Designing test objects that verify that the internal programme logic is operational properly and that programme input produces legitimate outputs is a component of unit testing. Internal code flow and all choice branches must be approved. It is the testing of various requests' software components.Before integration, it is done following the closure of each independent unit. This is an intrusive structural test that depends on knowledge of the object's structure. Unit tests do fundamental testing at the factor level and evaluate a particular business procedure, application, and/or system configuration. Unit tests make assurance that every step of a business process follows the written instructions precisely, has clearly specified inputs, and is likely to produce the expected outcomes.

8.2.2 FUNCTIONAL TESTING

Functional tests offer methodical verification that the tested functions are usable as indicated by the user manuals, system certification, and business and technical requirements. Functional difficulty is focused on the following things:

A class of valid input that must be received is identified using the term valid input.

The term "Invalid Input" refers to many categories of unauthorised input that must be rejected.

Functions must be utilised for the specified purposes.

Modules of request outputs are categorised using output.

Systems and procedures are used to interface with other systems or to appeal events.

Functional tests are organised and grounded with a focus on materials, essential functions, or unique test items. Additionally, thorough planning must go into evaluating the systematised coverage of data fields, specified procedures, and following processes as well as business process flows.

8.2.3 INTEGRATION TESTING

Integration tests are designed to evaluate if integrated software components truly operate as a single programme. Testing is event-driven and is primarily focused on the fundamental outcome of screens or fields. Integration tests confirm that, while unit testing successfully demonstrating that the workings were individually approved, the component grouping is accurate and dependable. The purpose of integration testing is to identify issues that result from the grouping of components.

8.2.4 WHITE BOX TESTING

White Box Testing is a challenging in which the software tester has information of the inner workings, construction and language of the software, or at least its drive. It is used to test areas that cannot be stretched from a black box level.

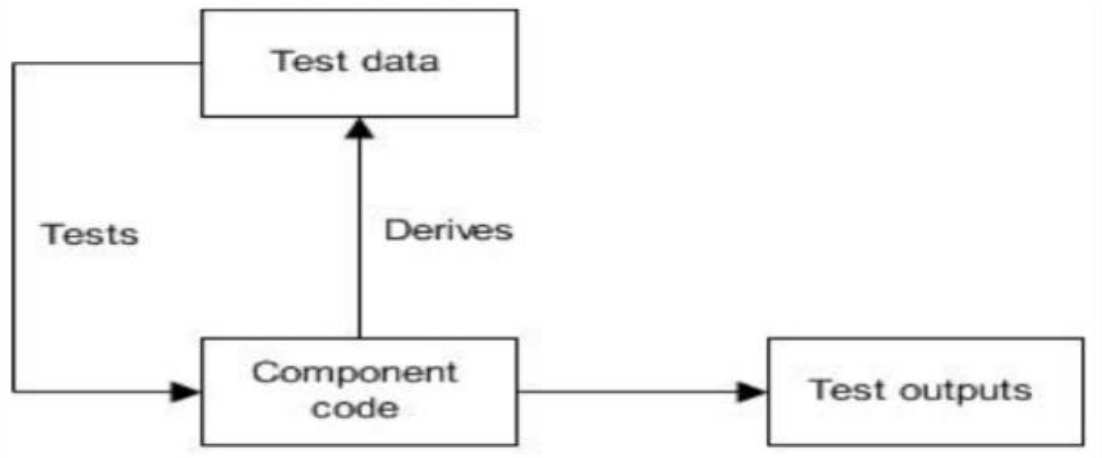


Fig.8.2.4 White Box Testing

8.2.5 BLACK BOX TESTING

Black Box Testing is testing the software short of any knowledge of the inner mechanisms, building or language of the module actuality tested. Black box tests, as most other kinds of tests, must be printed from a final source document, such as requirement or necessities file, such as specification requirement file. It is a testing in which the software below test is treated, as a black box. We cannot "see" into it. Equivalence Class Testing: It is used to minimize the number of possible test cases to an optimum level while maintains reasonable test coverage.

- Boundary Value Testing: Boundary value testing is focused on the values at boundaries. This technique determines whether a certain range of values are acceptable by the system or not. It is very useful in reducing the number of test cases. It is most suitable for the systems where an input is within certain ranges.
- Decision Table Testing: A decision table puts causes and their effects in a matrix. There is a unique combination in each column

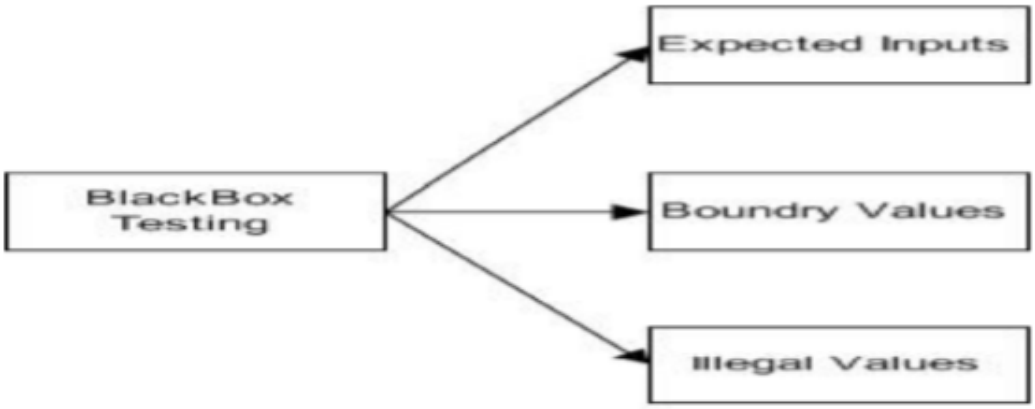


Fig.8.2.5 Black box Testing

8.2.6 SYSTEM TESTING

System testing confirms that the entire combined software system meets supplies. It tests a configuration to ensure known and predictable outcomes. An example of system testing is the arrangement-oriented system mixing test. System testing is based on procedure similes and flows, emphasizing pre-driven process links and addition points.

Testing the fully integrated applications including external peripherals in order to check how components interact with one another and with the system as a whole. This is also called End to End testing scenario.

Verify thorough testing of every input in the application to check for desired outputs.

Testing of the user's experience with the application.

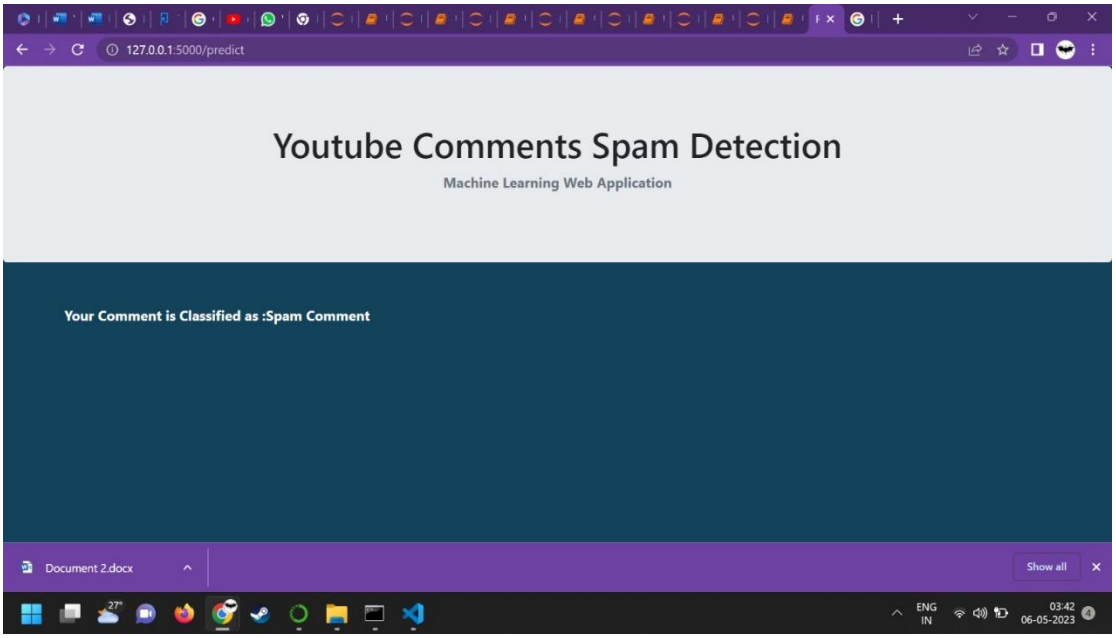
8.2.7 ALPHA TESTING

The alpha test in software development is a test conducted by the teams to ensure that your invention functions as intended. Alpha tests were originally used to refer to the first testing phase of a software development process. Component, module, and scheme testing are all covered in the first stage. Additionally, it enables us to evaluate the output on the last common denominator tackles to confirm that the preloads and transfer timings are appropriate. Without knowing how the software functions, the test gives inputs and responds to outputs.

8.2.8 BETA TESTING

A beta test is the second opinion of a software challenge in which a sample of the intended customers uses the product. Beta testing is a limited form of "prerelease testing." The database is currently being tested in "real-world" scenarios by curricular organisations and teachers using beta versions of the programme.

SCREENSHOTS AND OUTPUT



SOURCE CODE

```
from flask import Flask,render_template,url_for,request
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.feature_extraction.text import CountVectorizer
from sklearn.naive_bayes import MultinomialNB

app = Flask(_name_)

@app.route('/')
def home():
	return render_template('home.html')

@app.route('/predict', methods=['POST'])
def predict():
	df = pd.read_csv("YoutubeSpamMergedData.csv")
	df_data = df[["CONTENT", "CLASS"]]
	# Features and Labels
	df_x = df_data['CONTENT']
	df_y = df_data.CLASS
	# Extract Feature With CountVectorizer
	corpus = df_x
```

```
cv = CountVectorizer()
X = cv.fit_transform(corpus)  # Fit the Data

X_train, X_test, y_train, y_test = train_test_split(X, df_y, test_size=0.33,
random_state=42)
# Naive Bayes Classifier

clf = MultinomialNB()
clf.fit(X_train, y_train)
#clf.score(X_test, y_test)


if request.method == 'POST':
    comment = request.form['comment']
    data = [comment]
    vect = cv.transform(data).toarray()
    my_prediction = clf.predict(vect)
return render_template('result.html', prediction=my_prediction)



if _name_ == '_main_':
    app.run(debug=True)
```

```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.feature_extraction.text import CountVectorizer
from sklearn.naive_bayes import MultinomialNB

df = pd.read_csv("YoutubeSpamMergedData.csv")

df
```

out[3]:

	Unnamed: 0	Unnamed: 1	COMMENT_ID	AUTHOR	DATE	CONTENT	CLASS
0	Psy	0	LZQPQhLyRh80UYxNuaDWhIGQYNQ96IuCg-AYWqNPjpU	Julius NM	2013-11-07T06:20:48	Huh, anyway check out this you[tube] channel: ...	1
1	Psy	1	LZQPQhLyRh_C2cTtd9MvFRJedxydaVW-2sNg5Diuo4A	adam riyati	2013-11-07T12:37:15	Hey guys check out my new channel and our firs...	1
2	Psy	2	LZQPQhLyRh9MSZYnf8djyk0gEF9BHDPYrrK-qCczIY8	Evgeny Murashkin	2013-11-08T17:34:21	just for test I have to say murdev.com	1
3	Psy	3	z13jhp0bxqncu512g22wvzkasxmvvzjaz04	ElNino Melendez	2013-11-09T08:28:43	me shaking my sexy ass on my channel enjoy ^_^	1
4	Psy	4	z13fwbwp1oujthgqj04chlngpvzmtt3r3dw	GsMega	2013-11-10T16:05:38	watch?v=vtaRGgvGtWQ Check this out .	1
...	...	...	...	...	...	...	...
1951	Shakira	365	_2viQ_Qnc6-bMSjqyL1NKj57ROicCSJV5SwTrw-RFFA	Katie Mettam	2013-07-13T13:27:39.441000	I love this song because we sing it at Camp al...	0
1952	Shakira	366	_2viQ_Qnc6-pY-1yR6K2FhmC5i48-WuNx5CumlHLDAI	Sabina Pearson-Smith	2013-07-13T13:14:30.021000	I love this song for two reasons: 1.it is abou...	0
1953	Shakira	367	_2viQ_Qnc6_k_n_Bse9zVhJP8tJReZpo8uM2uZfnzDs	jeffrey jules	2013-07-13T12:09:31.188000	wow	0

1954 Shakira 368
_2viQ_Qnc6_yBt8UGMWyg3vh0PulTqcqyQtdE7d4Fl0 Aishlin Maciel
2013-07-13T11:17:52.308000 Shakira u are so wiredo 0
1955 Shakira 369
_2viQ_Qnc685RPw1aSa1tfrIuHXRvAQ2rPT9R06KTqA Latin Bosch
2013-07-12T22:33:27.916000 Shakira is the best dancer 0
1956 rows × 7 columns

```
df_data = df[["CONTENT", "CLASS"]]
df_x = df_data['CONTENT']
df_y = df_data.CLASS
import seaborn as sns
import matplotlib.pyplot as plt
%matplotlib inline
sns.countplot(df_data['CLASS'].values)
plt.xlabel('Class Values')
plt.ylabel('Class Counts')
plt.show()

corpus = df_x
cv = CountVectorizer()
X = cv.fit_transform(corpus)
X
out [8]: <1956x4454 sparse matrix of type '<class 'numpy.int64'>'
        with 25765 stored elements in Compressed Sparse Row format>
X_train, X_test, y_train, y_test = train_test_split(X, df_y, test_size=0.33, random_state=42)
clf = MultinomialNB()
clf.fit(X_train, y_train)
0ut[10]: MultinomialNB()
NAIVE_confidence = clf.score(X_test, y_test)
print("DecisionTree: ",NAIVE_confidence)
DecisionTree:  0.9195046439628483
```

CHAPTER 9

CONCLUSION

There are several methods used to categorise YouTube comments as spam and non spam (ham). This method has been put to the test using real-time YouTube comments, and the results show that it is 18% more accurate overall than the previous method. Since the YouTube API is a public platform accessible to all users, spammers' behaviour may evolve over time. The YouTube spam feature will not always remain the same in the real world; it frequently changes.

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